

## Session 2 Problems

### A. $3N + 1$

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Consider the following algorithm to generate a sequence of numbers.

Start with an integer  $n$ . If  $n$  is even, divide by 2. If  $n$  is odd, multiply by 3 and add 1. Repeat this process with the new value of  $n$ , terminating when  $n = 1$ . For example, the following sequence of numbers will be generated for  $n = 22$ :

22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1

Given  $n$ , print the generated sequence.

#### Input

The first line of input consists of a single integer  $T$  ( $1 \leq T \leq 100$ ), the number of test cases. Followed by  $T$  lines, each line consists of a single integer  $n$  ( $1 \leq n \leq 10^4$ ).

#### Output

For each test case, print the sequence generated by the integer  $n$ .

#### Sample test(s)

| input                                                          |
|----------------------------------------------------------------|
| 2<br>22<br>64                                                  |
| output                                                         |
| 22 11 34 17 52 26 13 40 20 10 5 16 8 4 2 1<br>64 32 16 8 4 2 1 |

## B. Egg Cartons

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

There are two types of egg cartons.

One type contains 6 eggs and the other type contains 8 eggs. John wants to buy exactly  $n$  eggs.

Help John find the minimum number of cartons he should buy.

### Input

The first line of input contains a single integer  $T$  ( $1 \leq T \leq 100$ ) the number of test cases. Followed by  $T$  lines, each line contains a single integer  $n$  ( $1 \leq n \leq 100$ ).

### Output

For each test case, print the minimum number of cartons John should buy. If there's no possible answer, print -1 instead.

### Sample test(s)

| input               |
|---------------------|
| 3<br>20<br>24<br>15 |
| output              |
| 3<br>3<br>-1        |

## C. Winnie the Pooh and honey

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

As we all know, Winnie-the-Pooh just adores honey. Ones he and the Piglet found out that the Rabbit has recently gotten hold of an impressive amount of this sweet and healthy snack. As you may guess, Winnie and the Piglet asked to come at the Rabbit's place.



Thus, there are  $n$  jars of honey lined up in front of Winnie-the-Pooh, jar number  $i$  contains  $a_i$  kilos of honey. Winnie-the-Pooh eats the honey like that: If the jar has less than  $k$  kilos of honey, he gives the jar to Piglet. Otherwise he eats it. Winnie does so until all jars are empty. Count how much honey Piglet will get.

### Input

The first line contains two integers  $n$  and  $k$  ( $1 \leq n \leq 100$ ,  $1 \leq k \leq 100$ ). The second line contains  $n$  integers  $a_1, a_2, \dots, a_n$ , separated by spaces ( $1 \leq a_i \leq 100$ ).

### Output

Print a single number — how many kilos of honey gets Piglet.

### Sample test(s)

|                               |
|-------------------------------|
| <b>input</b>                  |
| 3 20<br>10 20 30              |
| <b>output</b>                 |
| 10                            |
| <b>input</b>                  |
| 8 60<br>1 2 20 69 74 10 26 60 |
| <b>output</b>                 |
| 59                            |
| <b>input</b>                  |
| 1 100<br>300                  |
| <b>output</b>                 |
| 0                             |

## D. Not a Mysterious Machine

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Dr. Codenstein is at it again! Once again, Dr.Codenstein has invented a brilliant machine, but this time, he knows exactly what it does.

Dr.Codenstein is a big fan of Fibonacci, and he is obsessed with Fibonacci's sequence. Fibonacci's sequence is a sequence that defines its  $X$ th term as  $F(X) = F(X-1) + F(X-2)$ .

Given  $X$ , can you calculate the  $X$ th term?

### Input

The input consists of a single integer  $X$  ( $1 \leq X \leq 90$ ).

### Output

Your program should output the  $X$ th term in the Fibonacci sequence.

### Sample test(s)

|               |
|---------------|
| <b>input</b>  |
| 1             |
| <b>output</b> |
| 1             |
| <b>input</b>  |
| 4             |
| <b>output</b> |
| 3             |
| <b>input</b>  |
| 10            |
| <b>output</b> |
| 55            |

### Note

Note that  $F(1) = F(2) = 1$ .

## E. SoundCloud

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Kamal listens to music on SoundCloud alot, everywhere and at any time.

This annoys his friends. They say that he spends all his time listening to music and he quit doing anything else useful.

Kamal is bored from their advice to stop listening to music and decided to challenge them. If Kamal's friends win the challenge, Kamal will stop listening to music forever.

Kamal will start listening to his favorite playlist - which includes 20 songs, starting from song #1, but he made a mistake and told his friends that he will listen to N songs from the playlist (repetitions included).

Kamal's friends want to know the number of the last song Kamal listened to, after N Songs.

Help Kamal's friends win the challenge.

### Input

The input consist of N, the number of songs Kamal will listen to ( $1 \leq N \leq 1000$ ). Then N characters follow:

- '\*' means Kamal is listening to the current song.

- '>' means Kamal skipped the current song and will listen to the next song.

- '<' means Kamal repeated the last song.

The first character of the N number of characters will be either '\*' or '>'.

### Output

Print the number (in the playlist) of the last song Kamal has listened to after N songs.

### Sample test(s)

| input      |
|------------|
| 7<br>***** |
| output     |
| 7          |

| input       |
|-------------|
| 6<br>*>*>*> |
| output      |
| 9           |

### Note

- Kamal may play the playlist many times.

## F. Divisors

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Given a positive integer  $X$ , you are to find the number of divisors of  $X$ .

### Input

The first line of input contains  $T$  ( $1 \leq T \leq 100$ ), the number of test cases. Followed by  $T$  lines, each contains a single integer  $X$  ( $1 \leq X \leq 10^9$ ).

### Output

For each test case, print the number of divisors of  $X$ .

### Sample test(s)

| input              |
|--------------------|
| 3<br>12<br>7<br>36 |
| output             |
| 6<br>2<br>9        |

## G. Palindromic Numbers

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Gamil is writing a letter to his Indian pen-pal, but he isn't sure if Indians read numbers from left to right or from right to left. Gamil wants to make sure that any number he writes is read the same from both sides, to avoid any misunderstanding. You have a list with all the numbers Gamil will write in his letter. Can you tell Gamil how many of the numbers he will write may cause a misunderstanding?

### Input

The input consists of two lines, the first line consists of one integer  $N$  ( $1 \leq N \leq 100$ ), which the number of numbers you have on the list. The second line contains  $N$  numbers, separated by spaces, which are the numbers Gamil will write in his letter ( $0 \leq A_i \leq 10^{12}$ ).

### Output

Print one integer, which is how many numbers Gamil will write that may cause a misunderstanding.

### Sample test(s)

| input         |
|---------------|
| 3<br>121 1 35 |
| output        |
| 1             |

## H. Solution Set

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

You have been given an assignment but you are too lazy to do it, so you will write a program to do it for you.

Your program is meant to find all the possible solutions for the following equation:  $a^2 + b^2 + c^2 = d$  such that a, b, c and d are all non-negative integers.

### Input

You will be given an integer d. ( $0 \leq d \leq 10^5$ )

### Output

Print all the possible solutions for a, b and c, sorted lexicographically. (See the notes for more explanation.)

If there is no solution, print -1.

### Sample test(s)

| input                   |
|-------------------------|
| 1                       |
| output                  |
| 0 0 1<br>0 1 0<br>1 0 0 |

| input                   |
|-------------------------|
| 6                       |
| output                  |
| 1 1 2<br>1 2 1<br>2 1 1 |

### Note

Lexicographical order: [http://en.wikipedia.org/wiki/Lexicographical\\_order](http://en.wikipedia.org/wiki/Lexicographical_order)



# I. Carrots

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

John works at a company called "FIELD-Tech", and today, he was so tired after work that he fell asleep as soon as he got home. Unfortunately, even in his sleep, he was unable to forget about his work.

In one dream, he was asked to help a carrot producing company deal with the following question: how many carrots grow on a line segment connecting two given carrots? The endpoints of the segment (i.e., the two given carrots) should not be included.

It's a rather strange question, and to make it even stranger, the company's representatives (guys who have carrots instead of heads) said that all the carrots grow on an infinite plane, and there is exactly one carrot at each point with integer coordinates.

You must help tired John deal with this problem.

## Input

The first line of input contains a single integer  $T$  ( $1 \leq T \leq 200$ ), the number of test cases. Followed by  $T$  lines, each line contains 4 integers  $X_1, Y_1, X_2, Y_2$  ( $0 \leq X_1, Y_1, X_2, Y_2 \leq 50$ ), representing the coordinates of the two endpoints of the segment.

## Output

For each test case, print the number of carrots.

## Sample test(s)

| input                                             |
|---------------------------------------------------|
| 4<br>1 1 5 5<br>0 0 1 1<br>50 48 0 0<br>0 0 42 36 |
| output                                            |
| 3<br>0<br>1<br>5                                  |

# J. Leap Age

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Your friend was born on "leap day" and has always been troubled by the fact that he just turned 24 this year on his 5th birthday. You would like to write him a nice program for his birthday to help him and other leap-day-born people keep track of how many birthdays they've had so far.

## Input

The first line of input contains a single integer  $T$  ( $1 \leq T \leq 100$ ), the number of test cases. For each test case you will be given two integers  $A$  and  $B$  between 1582 and 10000,  $A$  represents the current year and  $B$  represents the year the person was born in ( $B \leq A$ ).

## Output

For each test case, print the number of leap days that have occurred in that time span, not including the first year if it is a leap year, but including the current year if it's a leap year.

## Sample test(s)

| input                                    |
|------------------------------------------|
| 3<br>2004 1980<br>2007 1981<br>1981 1980 |
| output                                   |
| 6<br>6<br>0                              |

## Note

-A leap year is any year that is a multiple of 4, unless it is divisible by 100 and not 400. For example, 1984 was a leap year, but 1900 wasn't, because it was divisible by 100 and not 400. 2000 was a leap year, because it was divisible by both 100 and 400.